



NUST CHIP DESIGN CENTRE

RISC-V Assembly

Lab Manual # 13

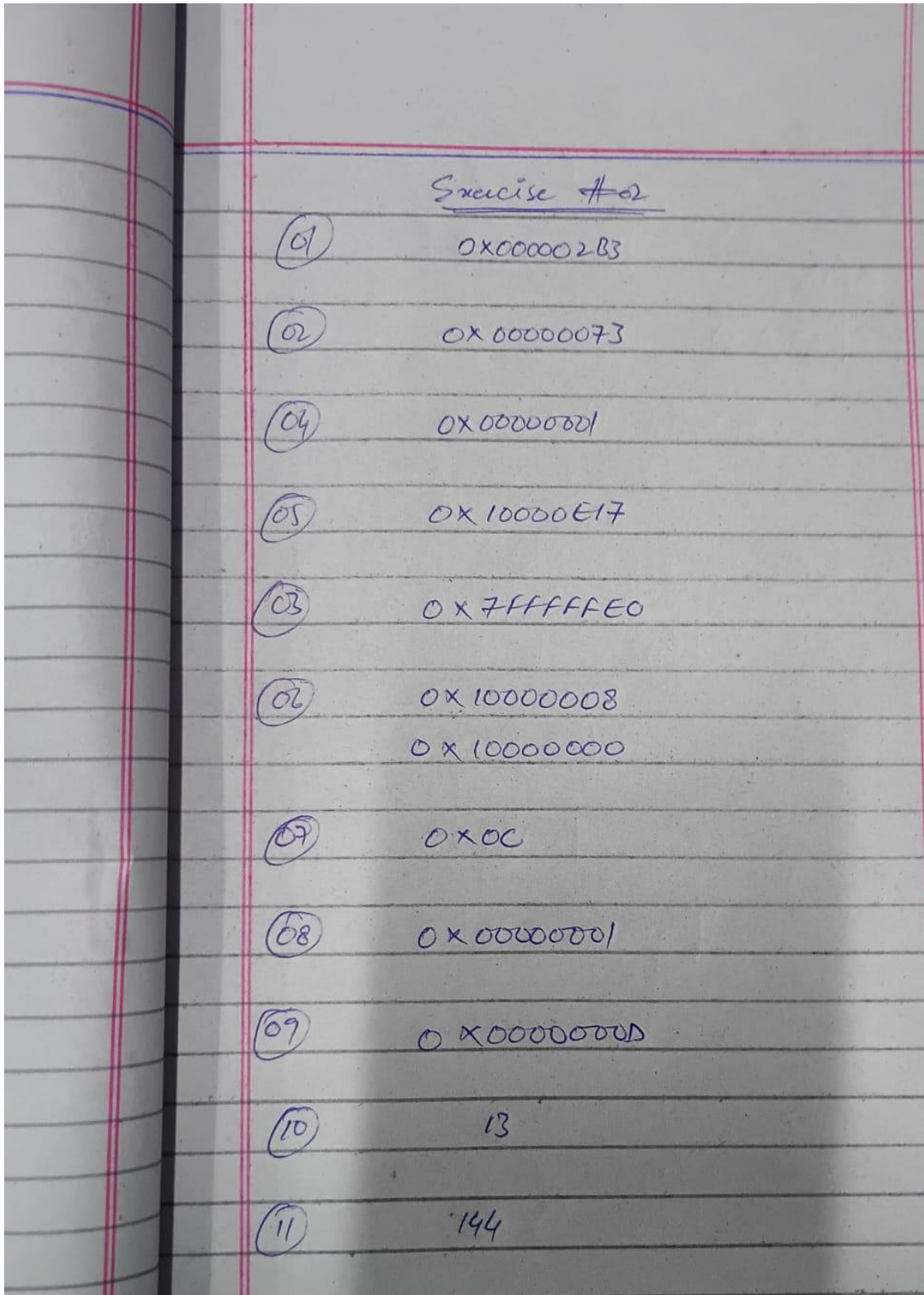
Data Transfer, Decision Making, Logical Ops

(Arrays in RISC-V Assembly, Bit Manipulation and Implementation of factorial function)

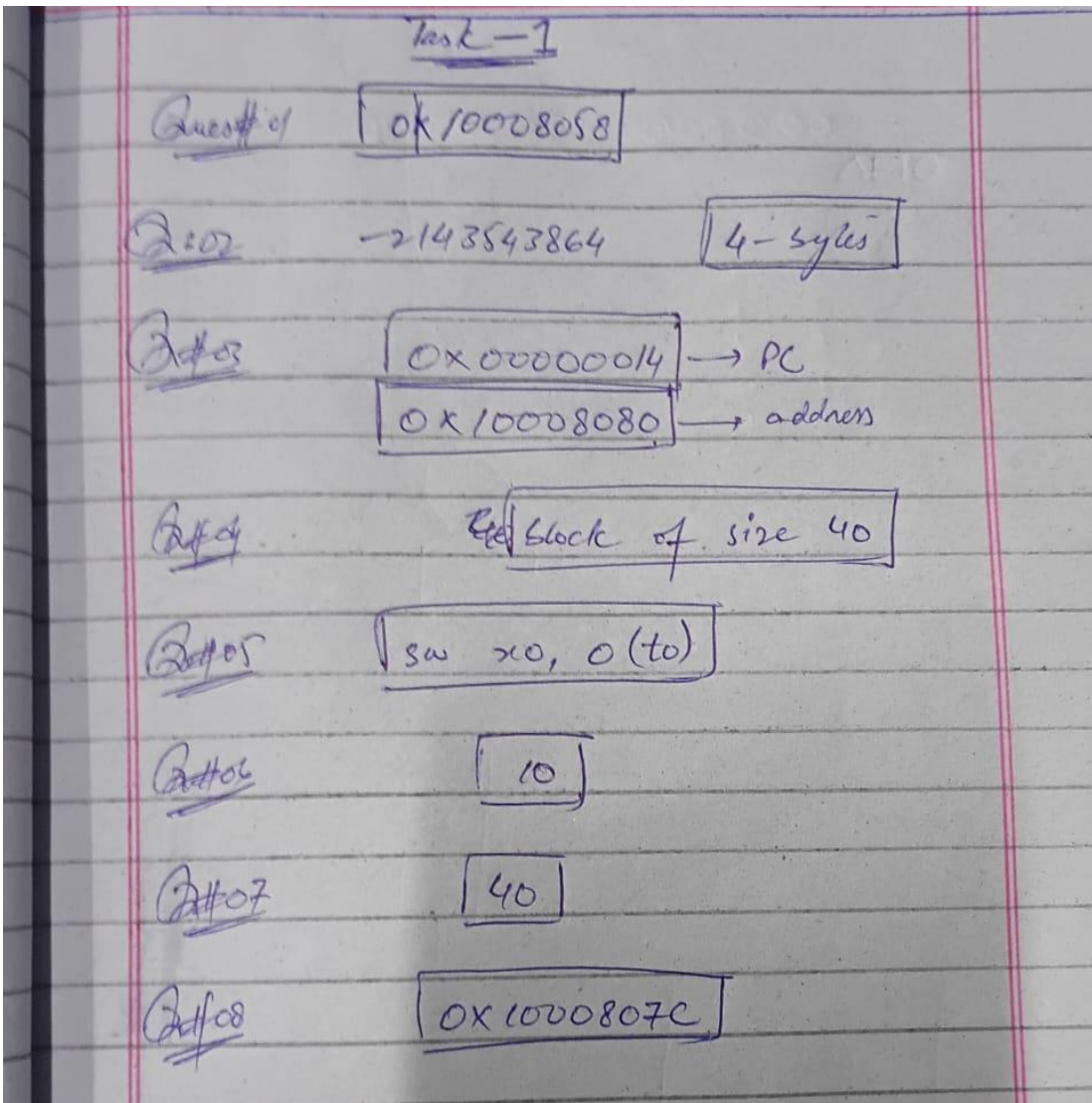
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<u>Date</u>	<i><u>13th Aug 2025</u></i>

1. In-Lab Tasks: (Write your lab task & screenshots here)

i. Exercise 2:



ii. Task 1:



```
[memcheck] malloc: editor.S:40 ecall
[memcheck] malloc: ptr=0x10008058 rawptr=0x10008030 size=40
[memcheck] heap allocs
[memcheck] ptr=0x10008058 size=40
[memcheck] end heap allocs
[memcheck] heap frees
[memcheck] end heap frees
```

```
[memcheck] access: addr=0x1000807C size=4 pc=0x00000014 file=editor.S:19 instr=sw x0 0(t0)

Exited with error code 0
[memcheck] In use at exit: 40 bytes in 1 blocks
40 bytes at 0x10008058 is lost
```

```
[memcheck] malloc: editor.S:40 ecall
[memcheck] malloc: ptr=0x10008058 rawptr=0x10008030 size=40
[memcheck] heap allocs
[memcheck]      ptr=0x10008058 size=40
[memcheck] end heap allocs
[memcheck] heap frees
[memcheck] end heap frees
```

```
[memcheck]      ptr=0x10008058 size=40
[memcheck] end heap frees

Exited with error code 0
[memcheck] In use at exit: 0 bytes in 0 blocks
```

iii. Task 2:

```
f(-3) should be 6, and it is: 6
f(-2) should be 61, and it is: 61
f(-1) should be 17, and it is: 17
f(0) should be -38, and it is: -38
f(1) should be 19, and it is: 19
f(2) should be 42, and it is: 42
f(3) should be 5, and it is: 5
```

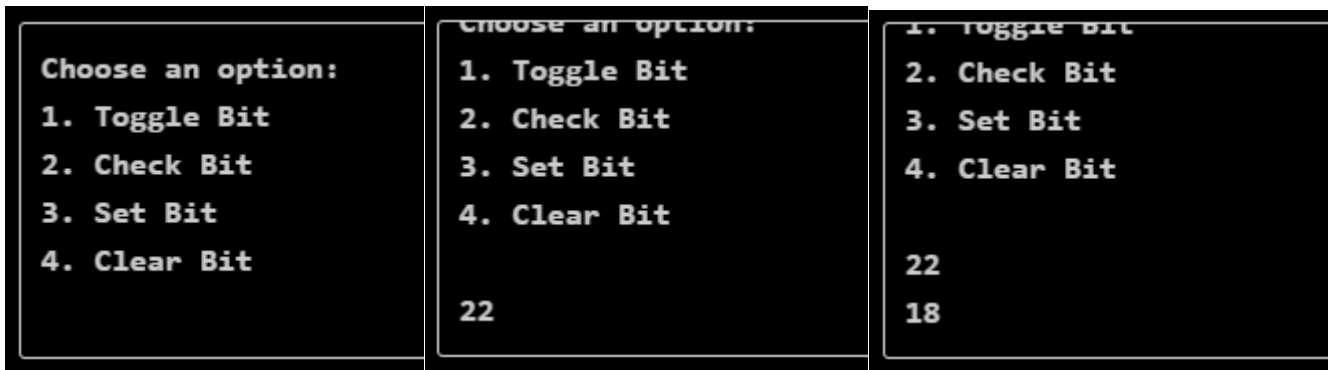
iv. Task 3:

```
-1 5 4 2 1 8 78 11 35 22
```

v. Task 4:

```
40320
```

vi. Task 5:



2. Critical Analysis: (Write you critical analysis / conclusion here)

In these lab tasks, the focus was on developing a deeper understanding of RISC-V assembly programming through progressively challenging exercises. Each task introduced new concepts such as analyzing assembly structure, working with arrays, handling data display, and performing bit-level operations. Collectively, they emphasized how low-level programming requires precise control of registers, memory, and instructions to achieve even simple outcomes. By practicing these different aspects, the labs provided a well-rounded foundation in assembly language, highlighting the importance of efficiency, accuracy, and step-by-step problem solving in system-level programming.